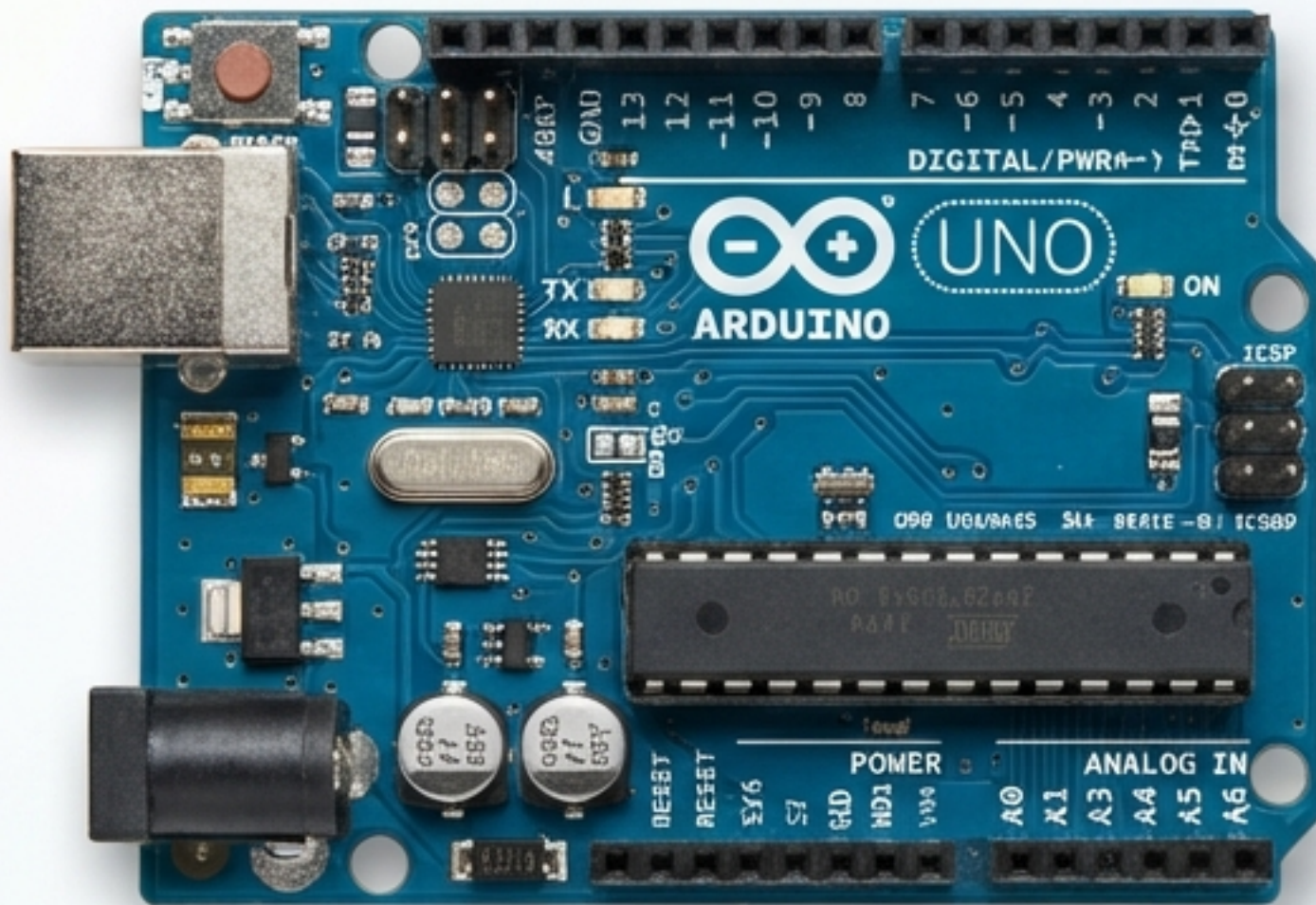
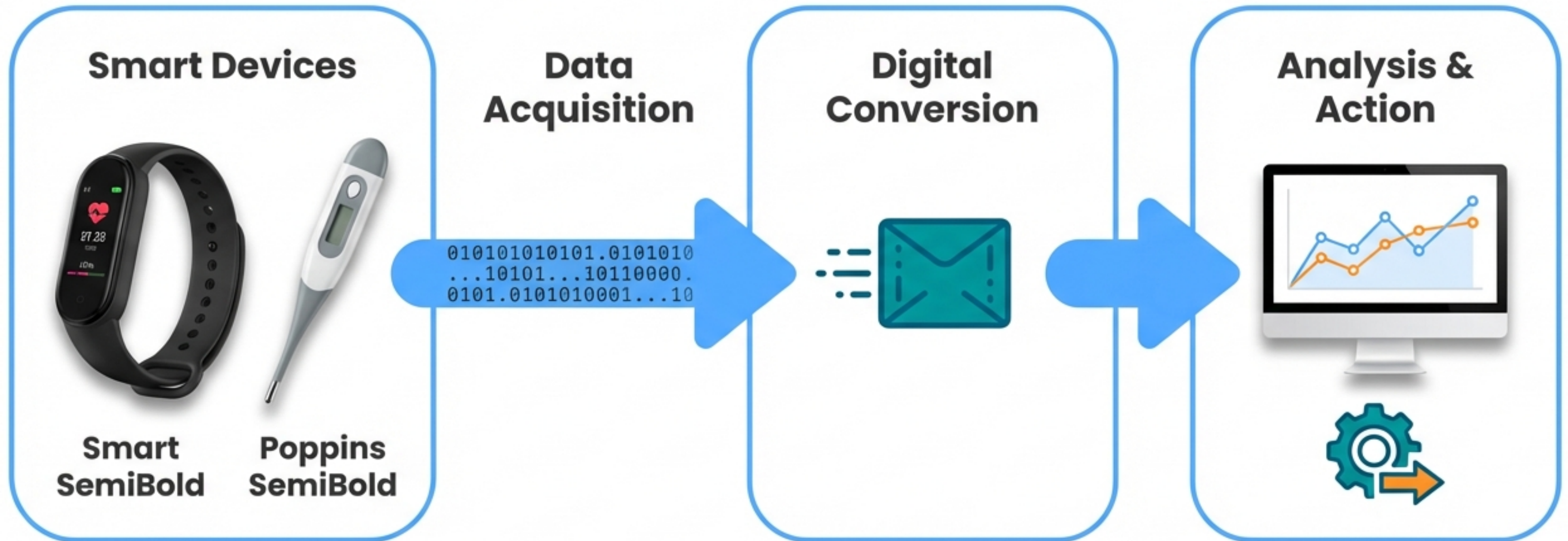


Arduino Programming: Building Smart Devices

Grade 7 - ICT 360

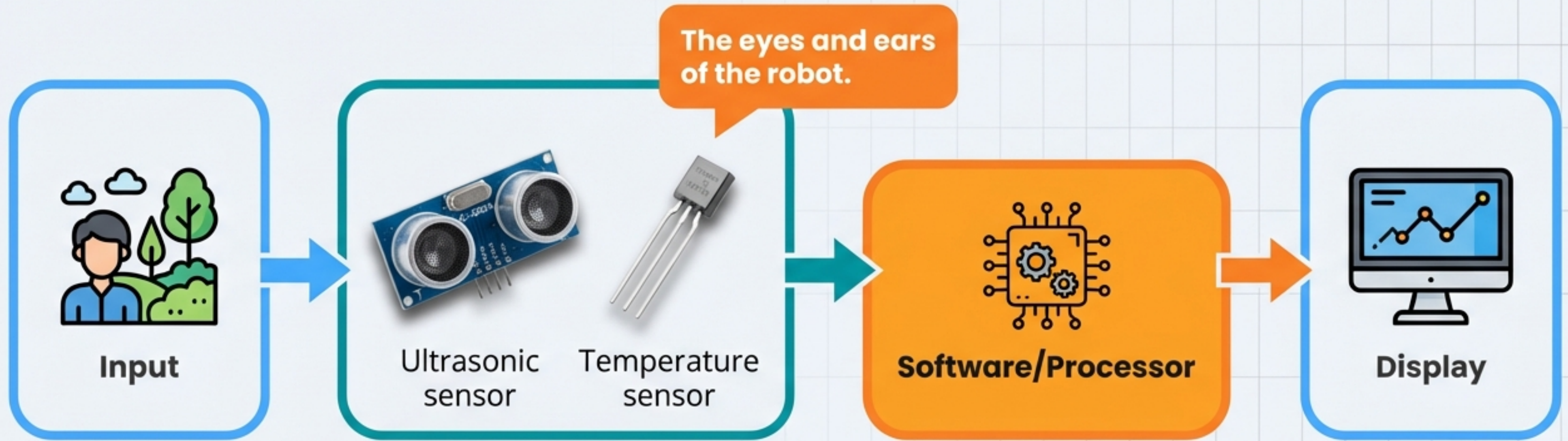


Data is the New Oil



Smart devices collect data (numbers, words, images) → Convert to digital
→ Analyze → Take Action.

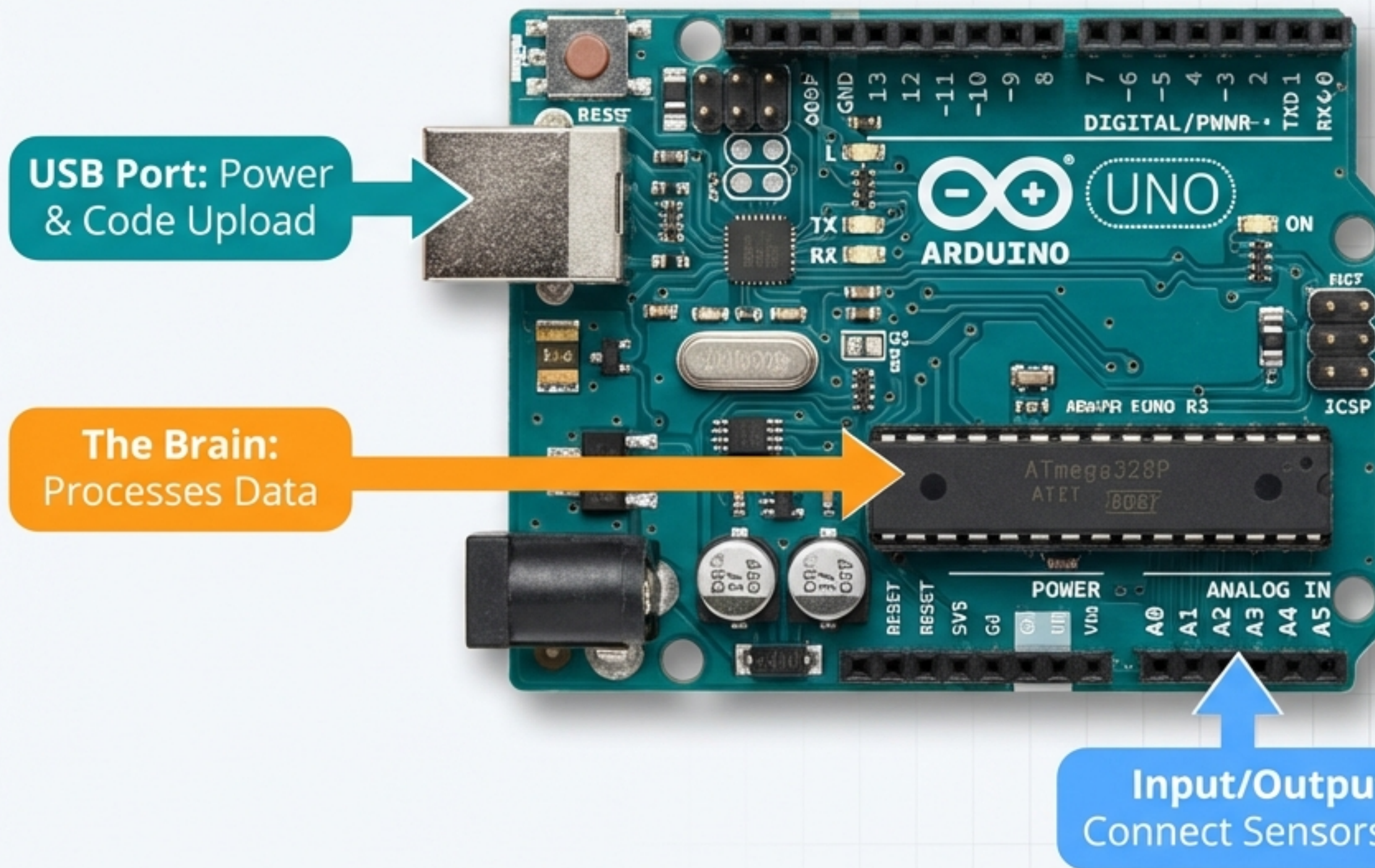
How Computers “Feel” the World



Sensors

Electronic devices that detect changes (like temperature or motion) and convert them into electronic signals.

Meet the Brain: Arduino Uno



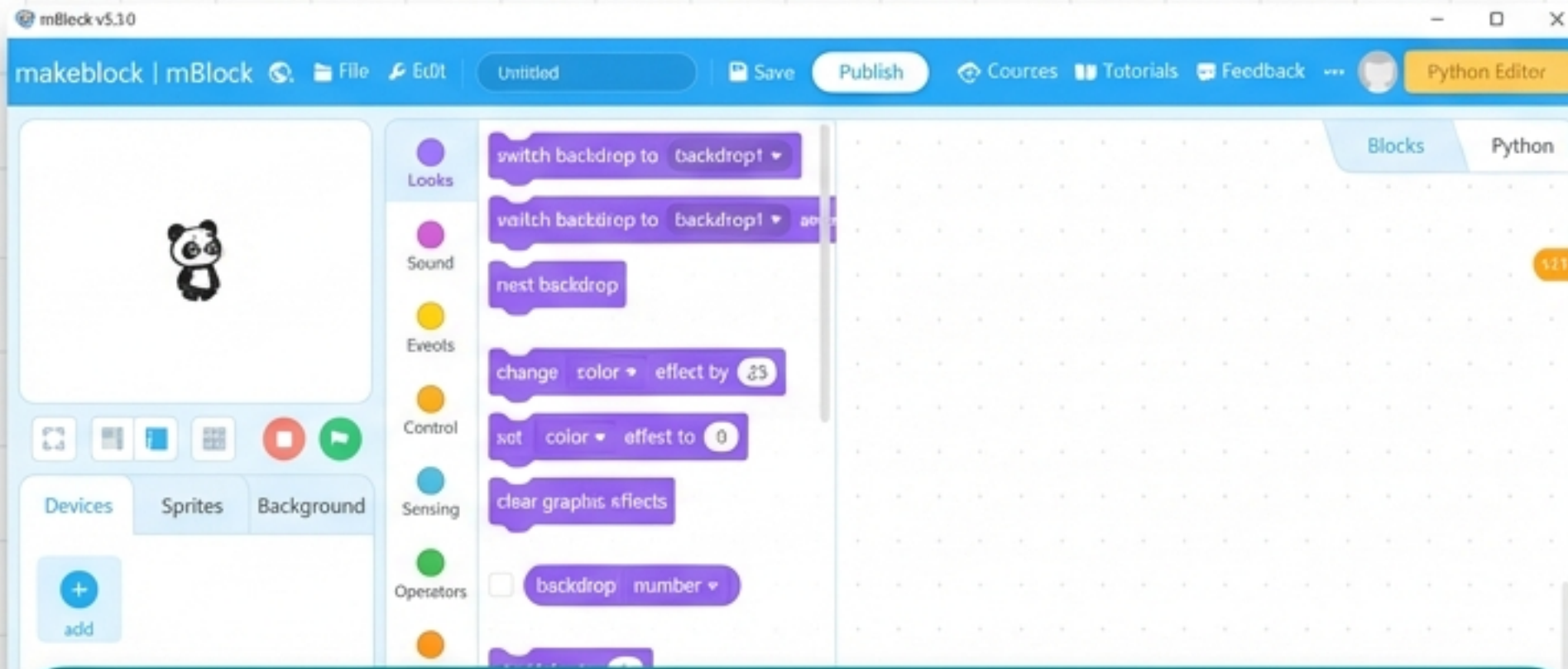
Key Roles

- ⚙️ **The Mediator:** Connects sensors to the computer.
- ⚙️ **Open-Source:** Easy hardware & software platform.
- ⚙️ **Input/Output:** Reads the world (Input) to control the world (Output).

ICT
360⁺

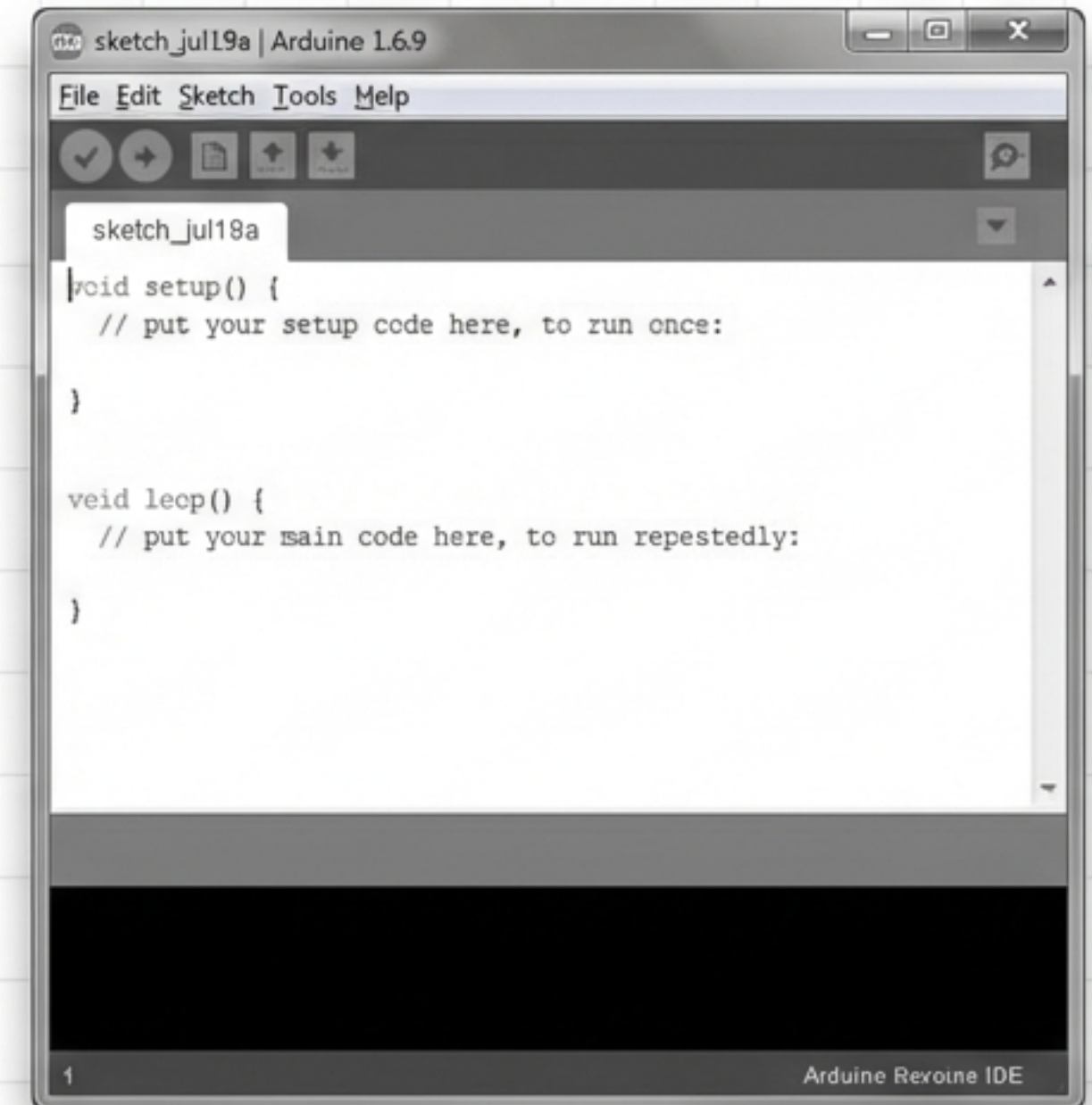
Speaking the Language: mBlock

Block-Based Coding (Easy!)

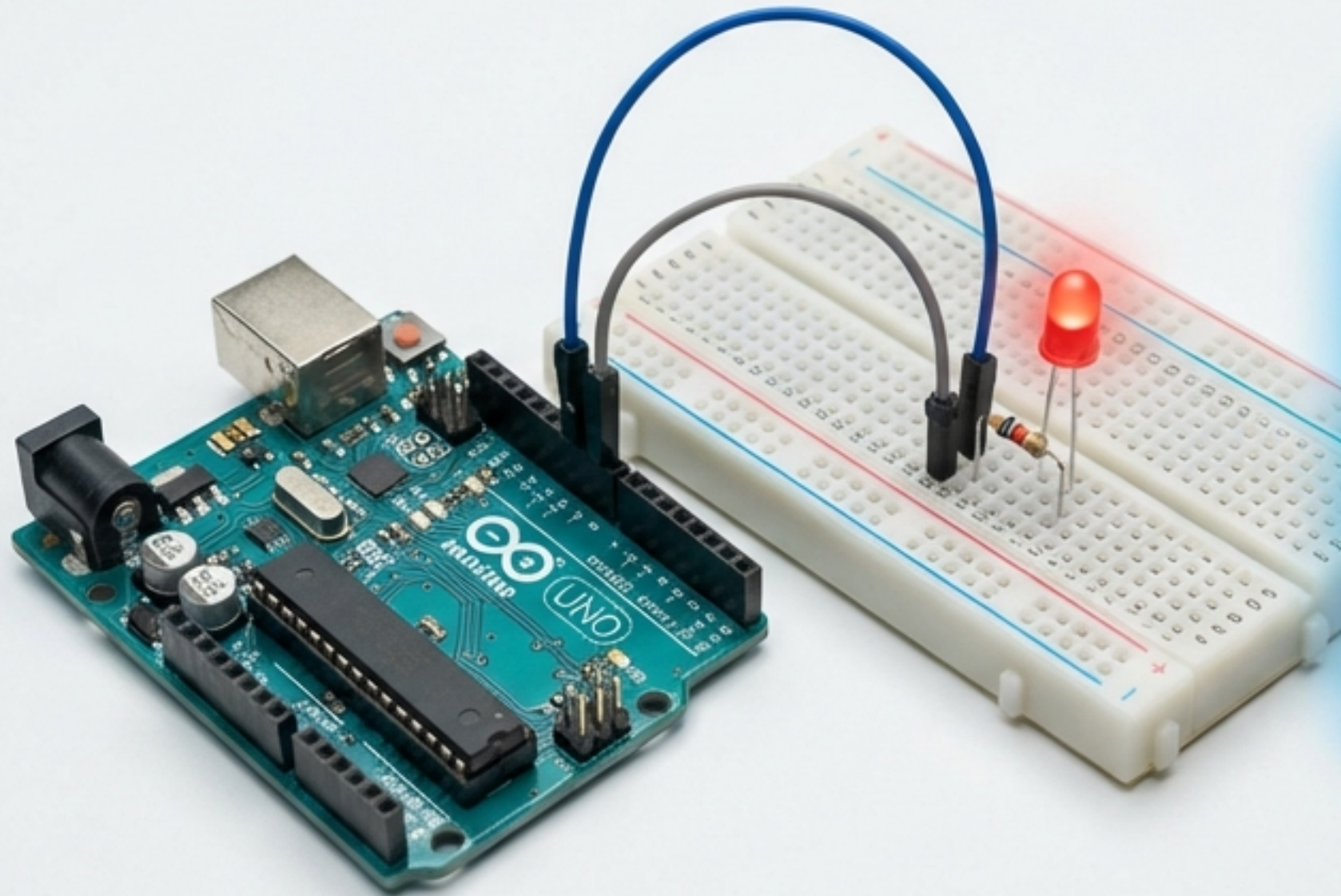


- **Visual Coding:** Like connecting puzzle pieces.
- **Simple Logic:** If this, then that.
- **Versatile:** Switch between blocks and text anytime.

Text-Based Coding



Mission: The Blinking LED Project



Goal: Create a stand-alone circuit that turns an LED on and off automatically.

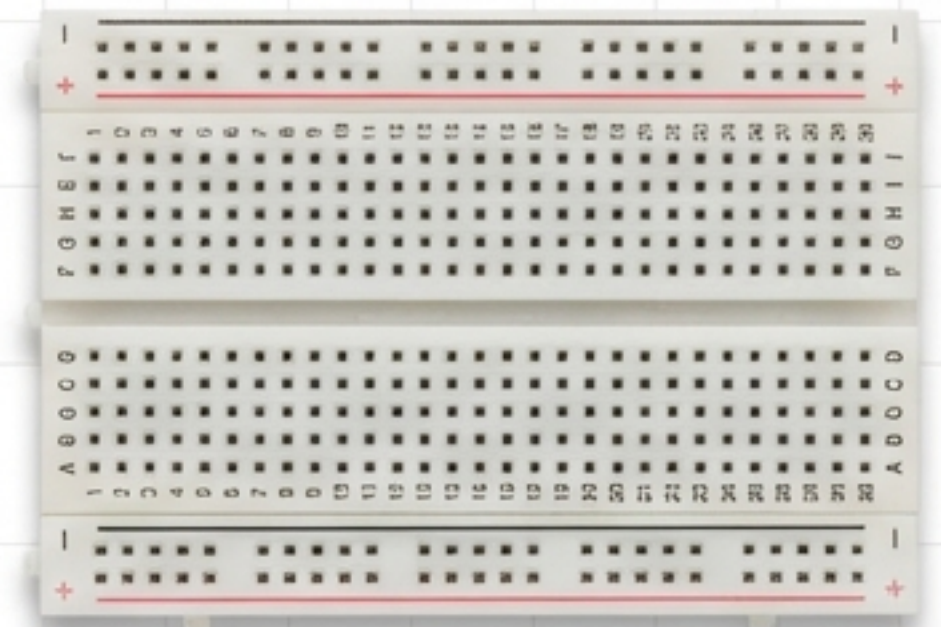
Objective: Control hardware with software logic.

Mission Inventory

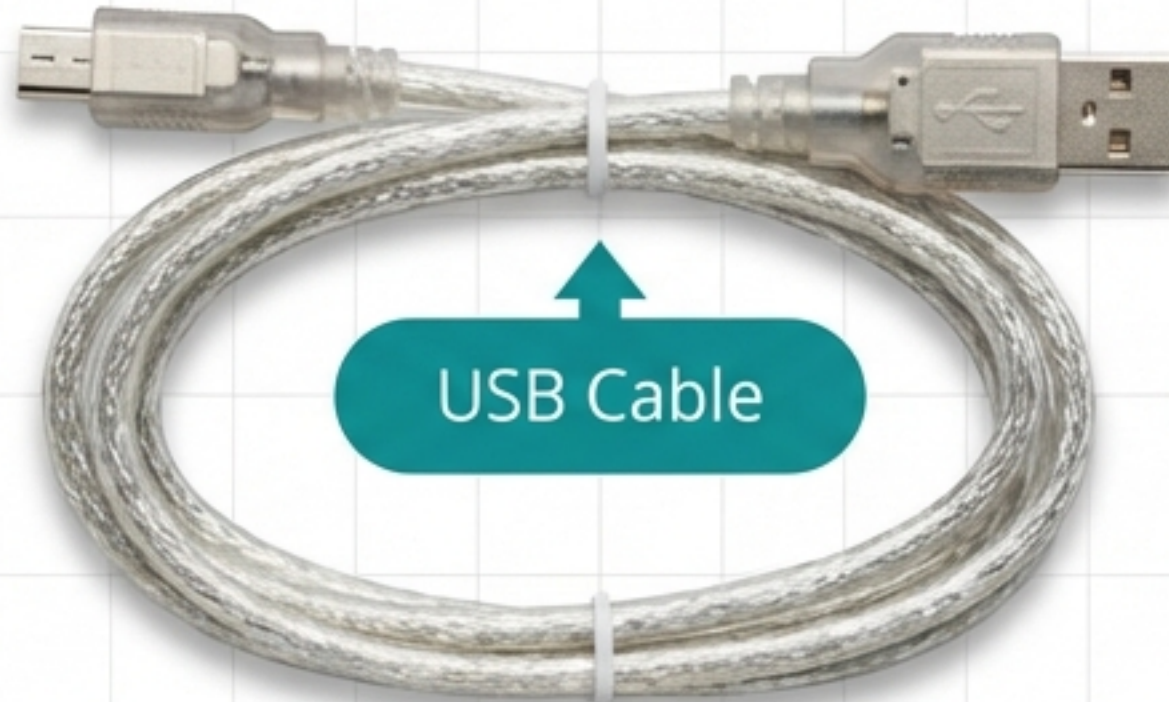
Arduino Uno Board



Breadboard



USB Cable



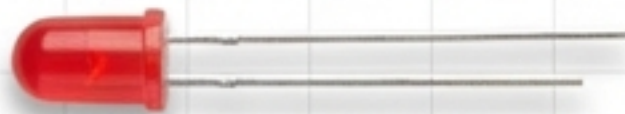
Resistor (470 ohms)



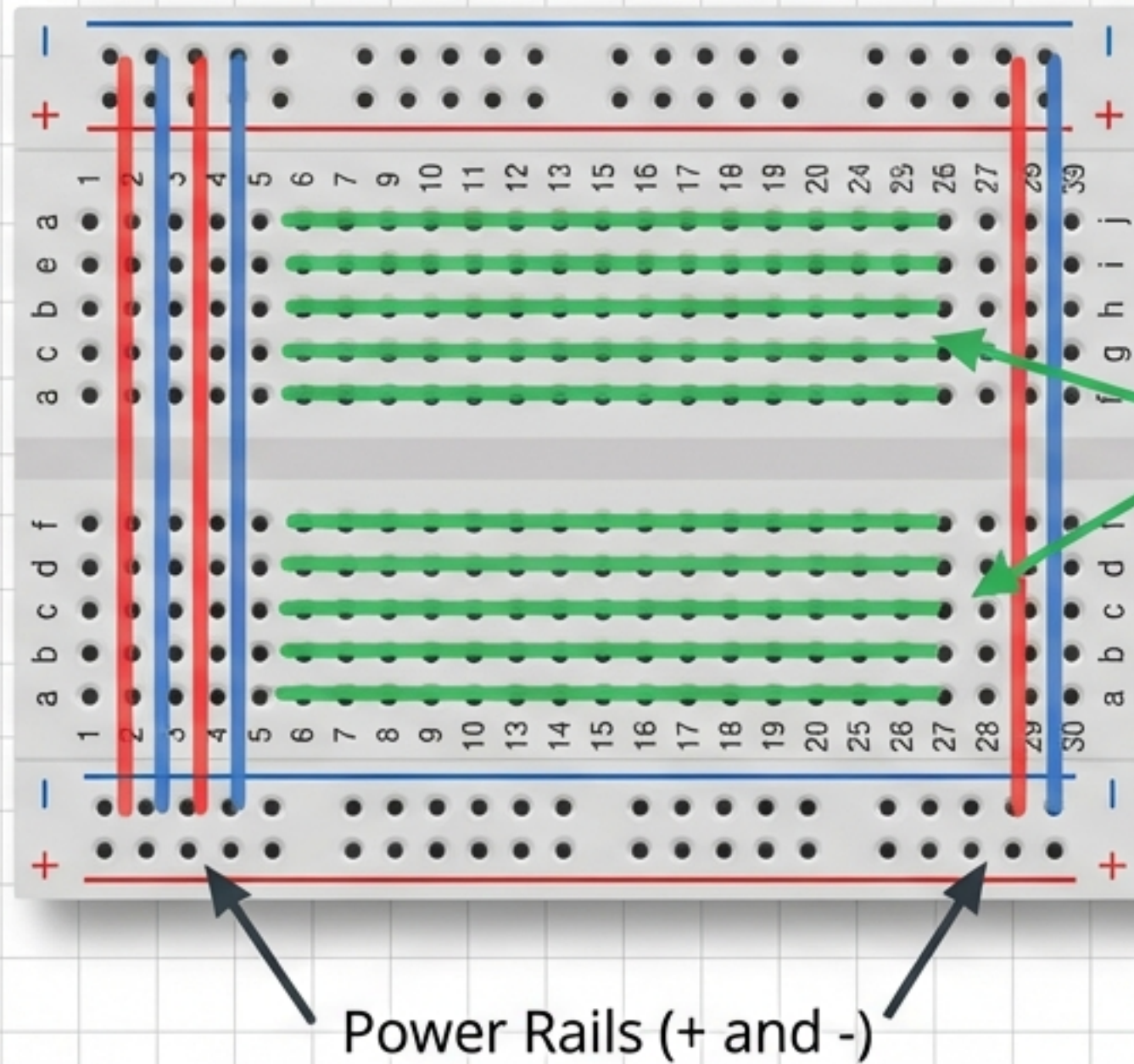
Jumper Wires



LED (Light Emitting Diode)



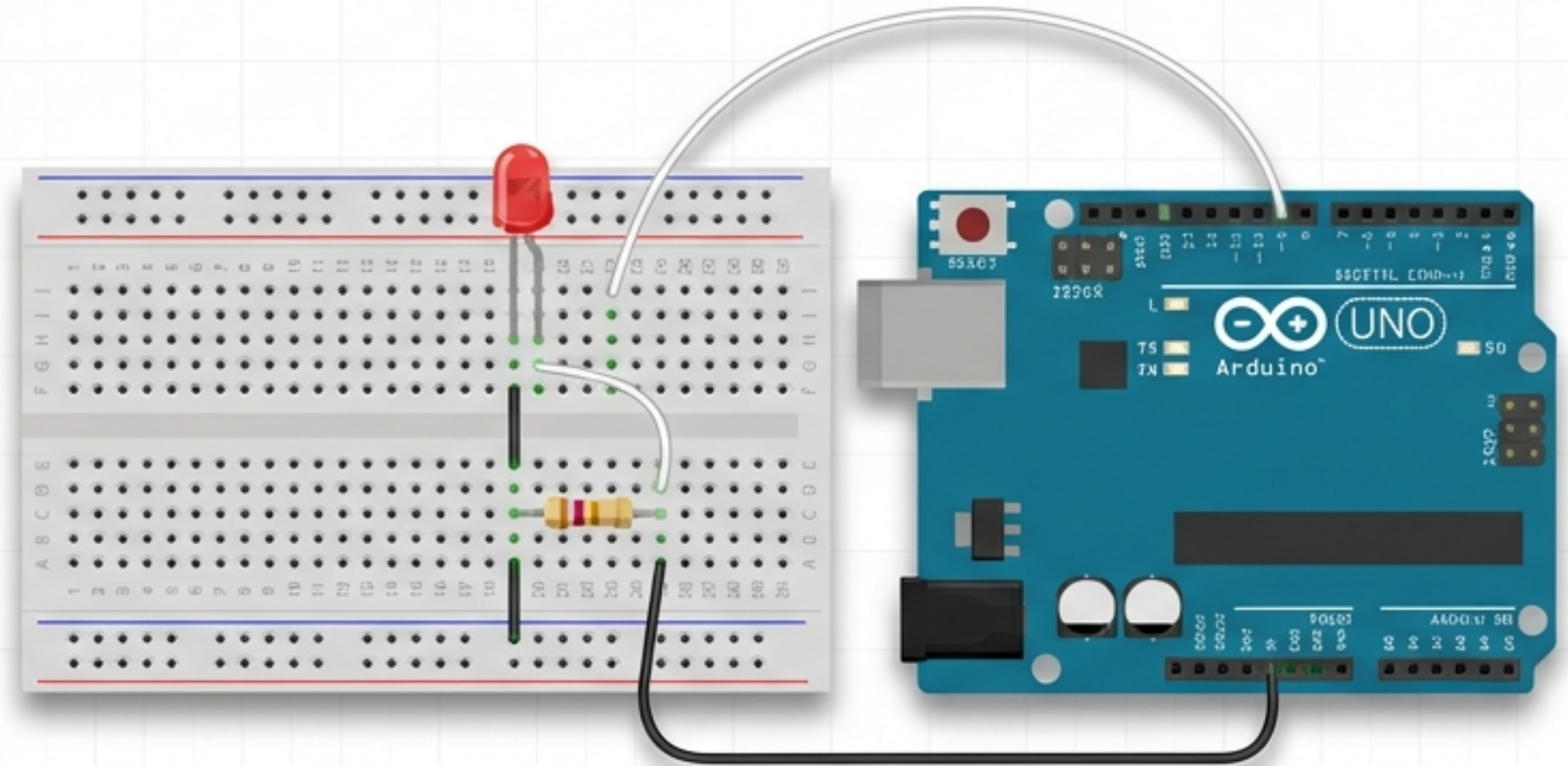
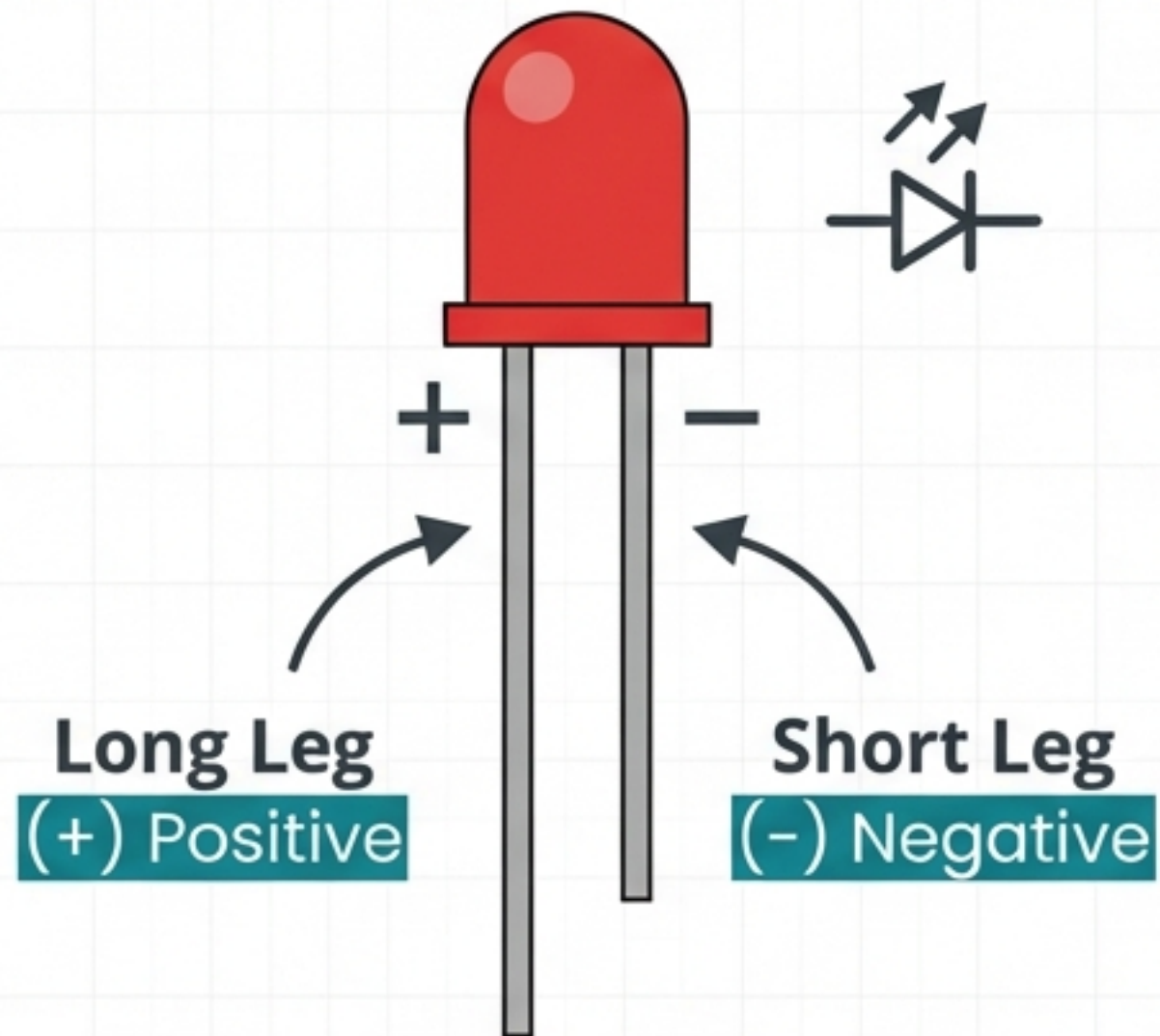
Understanding the Breadboard



Component Area
(Rows connected
horizontally)

The breadboard allows us to build circuits without soldering.

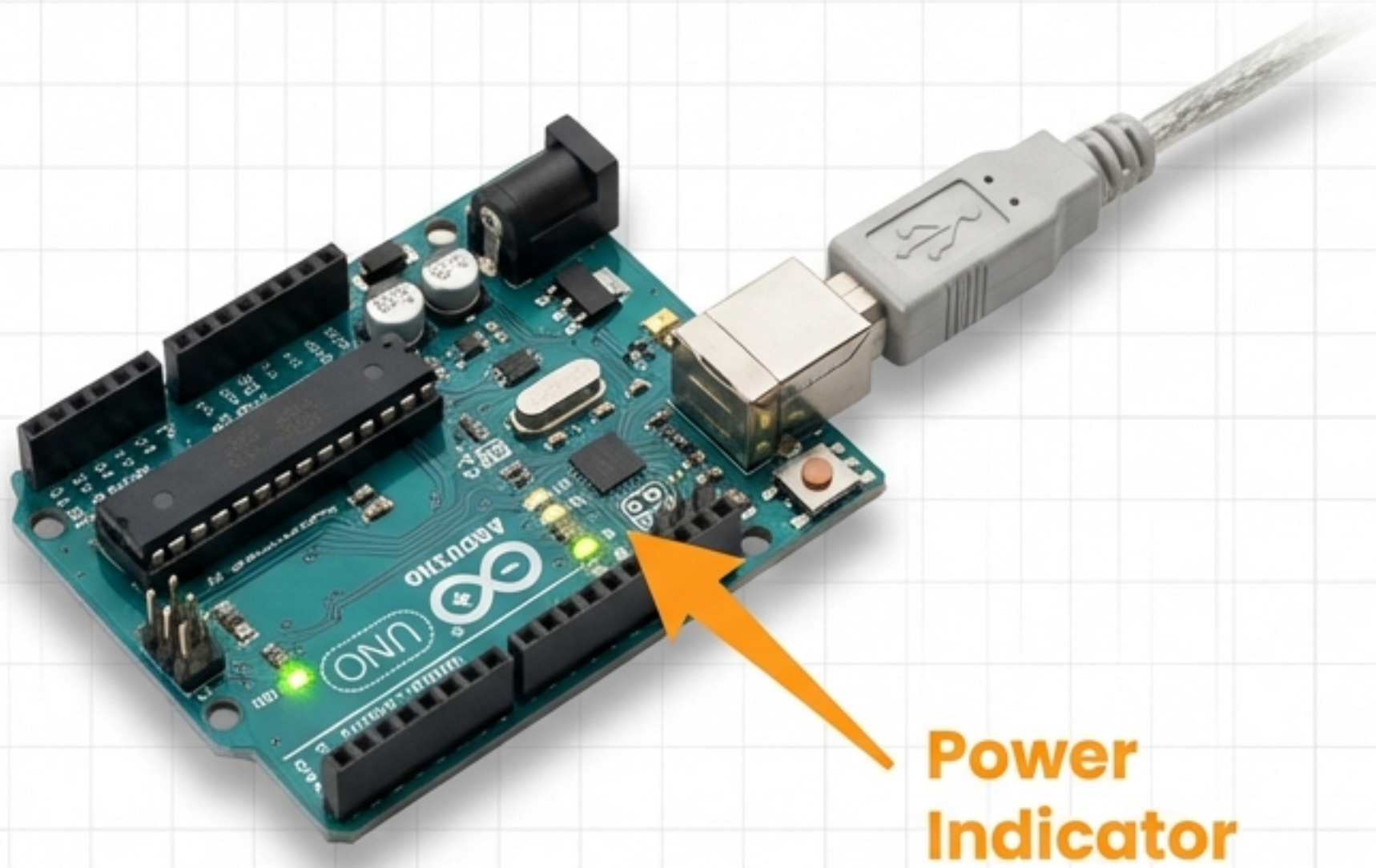
Wiring Step 1: The LED Circuit



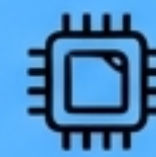
1. Connect LED **Negative** (Short Leg) to **GND** on Arduino.
2. Connect LED **Positive** (Long Leg) to Pin 8 via the **470Ω Resistor**.

⚠ Safety: The resistor protects the LED from burning out!

Wiring Step 2: Power Up

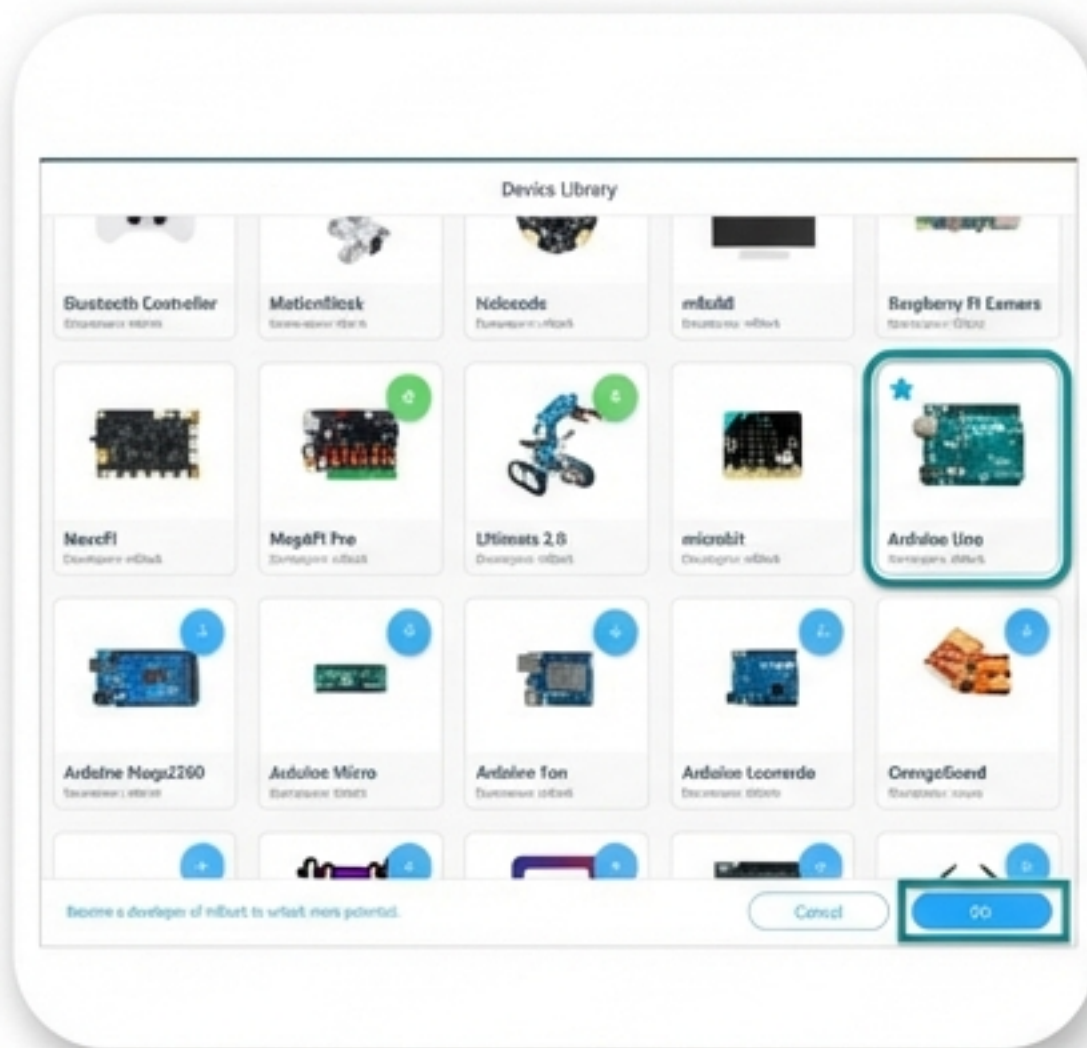


1. Plug the USB cable into the Arduino and your Computer.
2. Check: Ensure the Power LED on the board turns ON.



Tech Note: We are using Digital Pin 8 for our project.

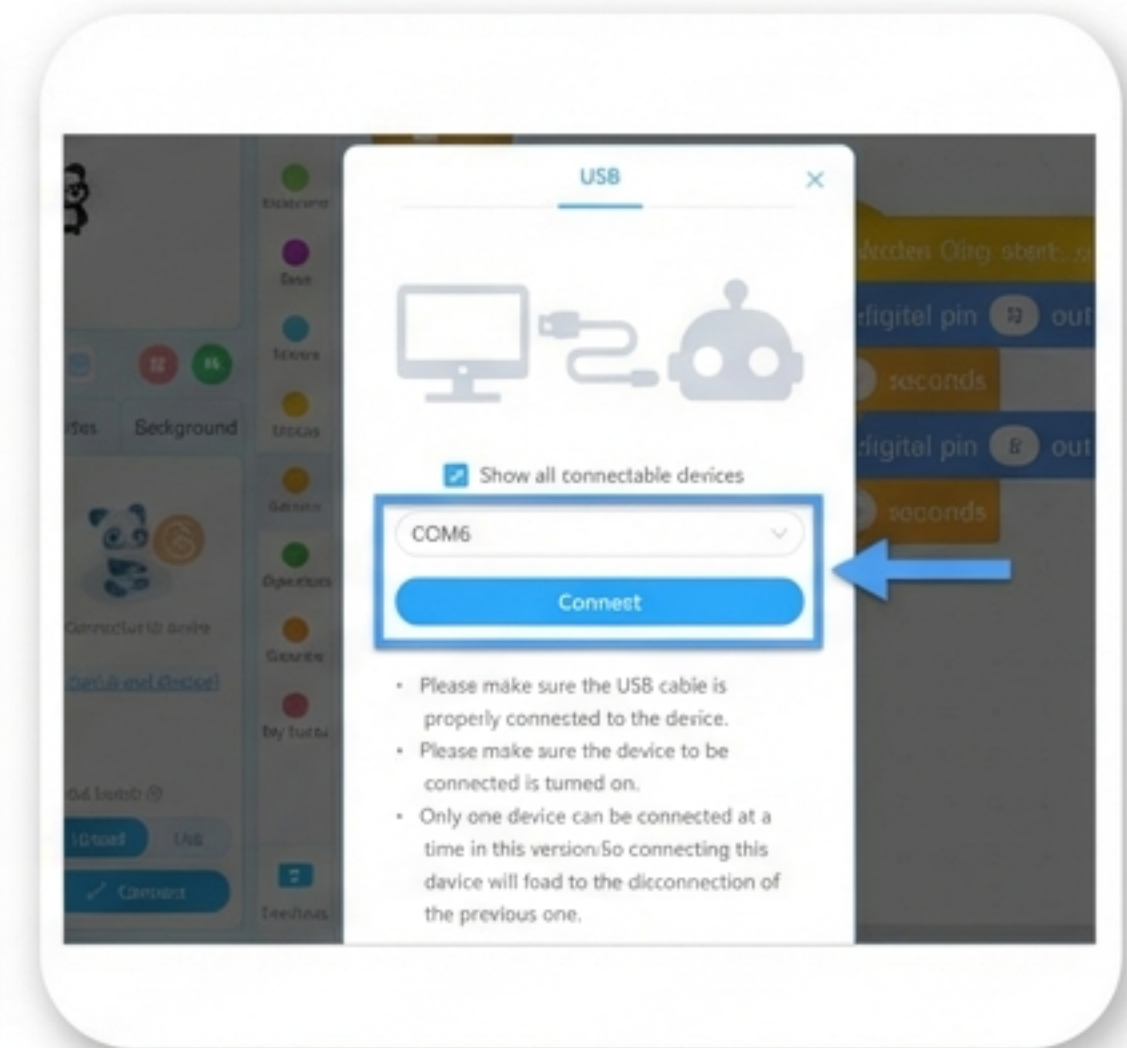
Setup: Connecting mBlock



1. Open mBlock. Click '**Add**' and select '**Arduino Uno**'.

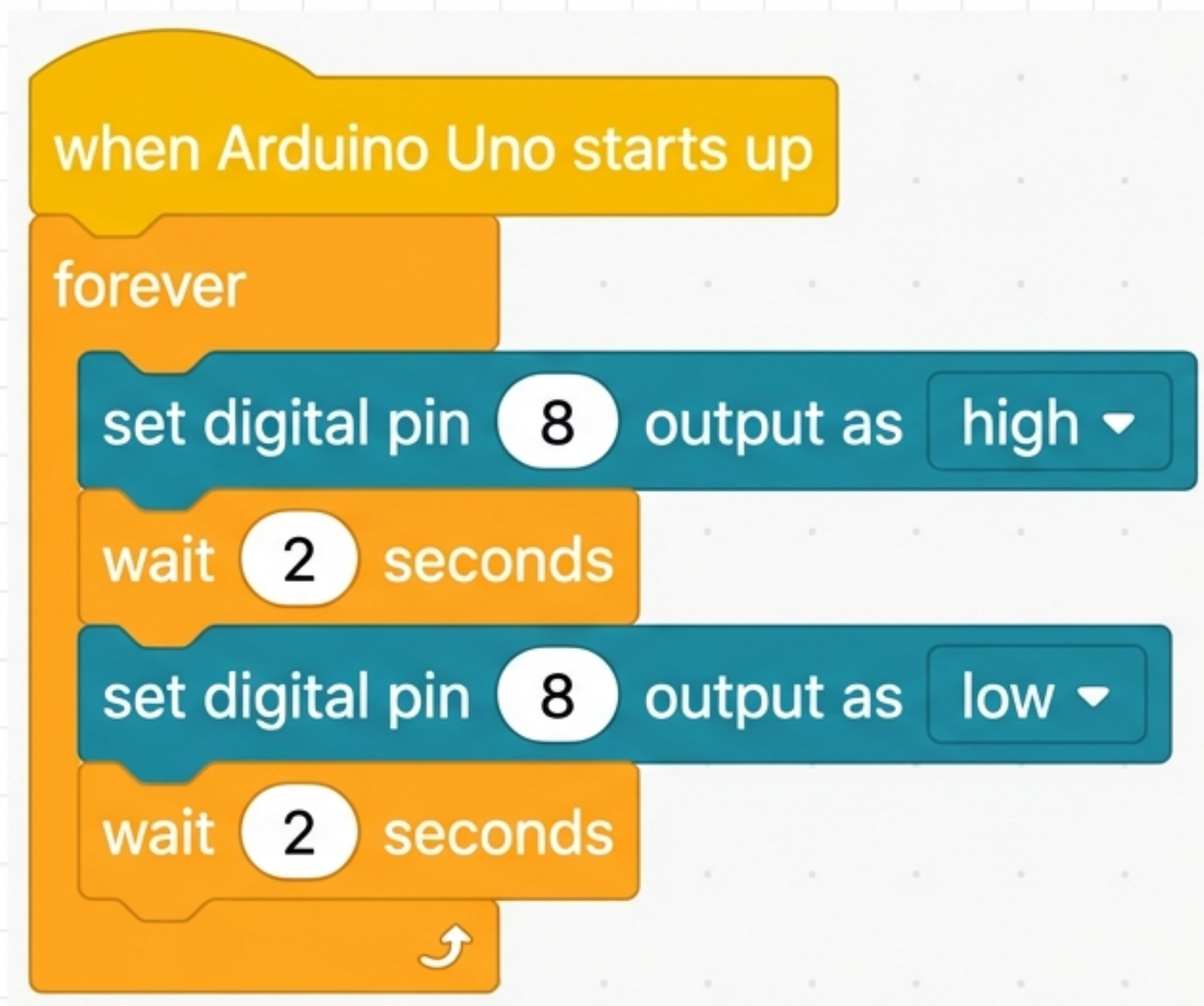


2. Click '**Connect**'.



3. Select your COM Port and wait for the 'Connected' message.

The Code: Digital Logic



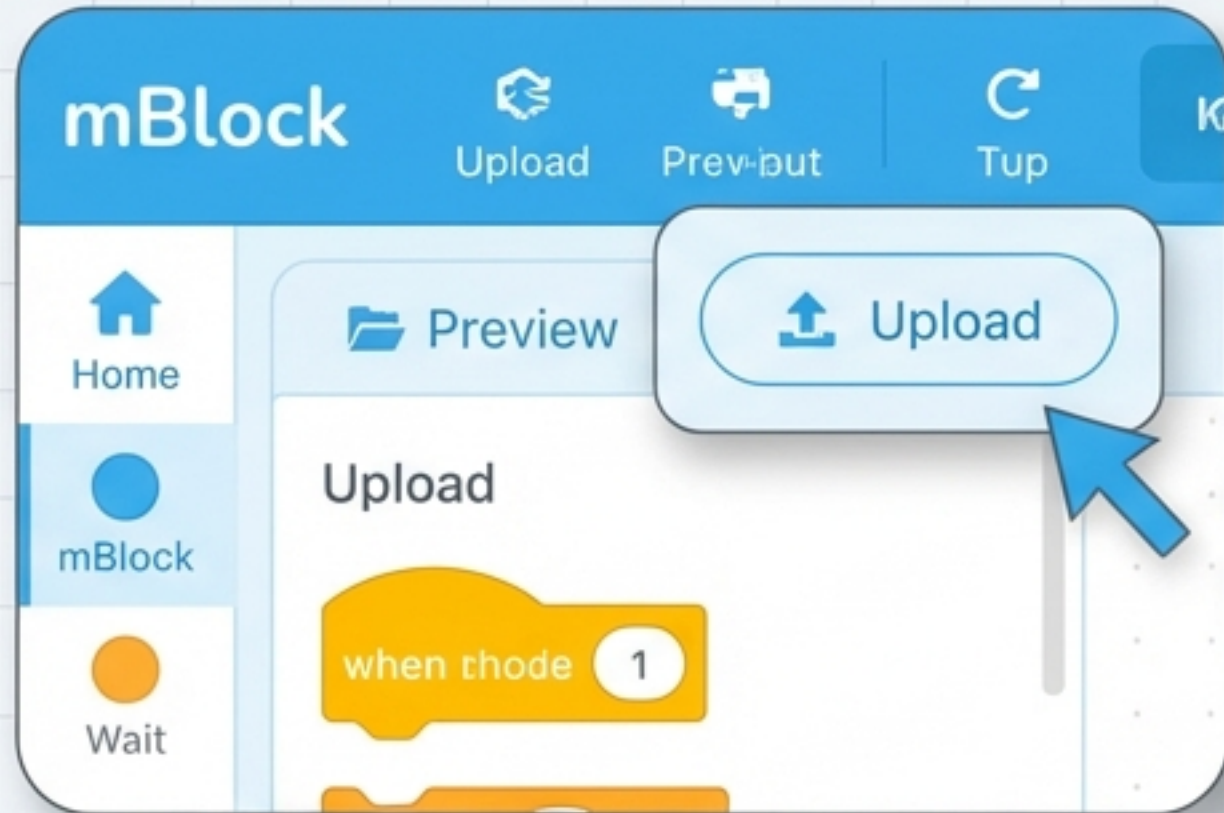
The Event (Start)

The Loop (Repeat Forever)

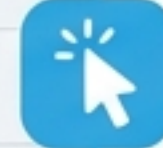
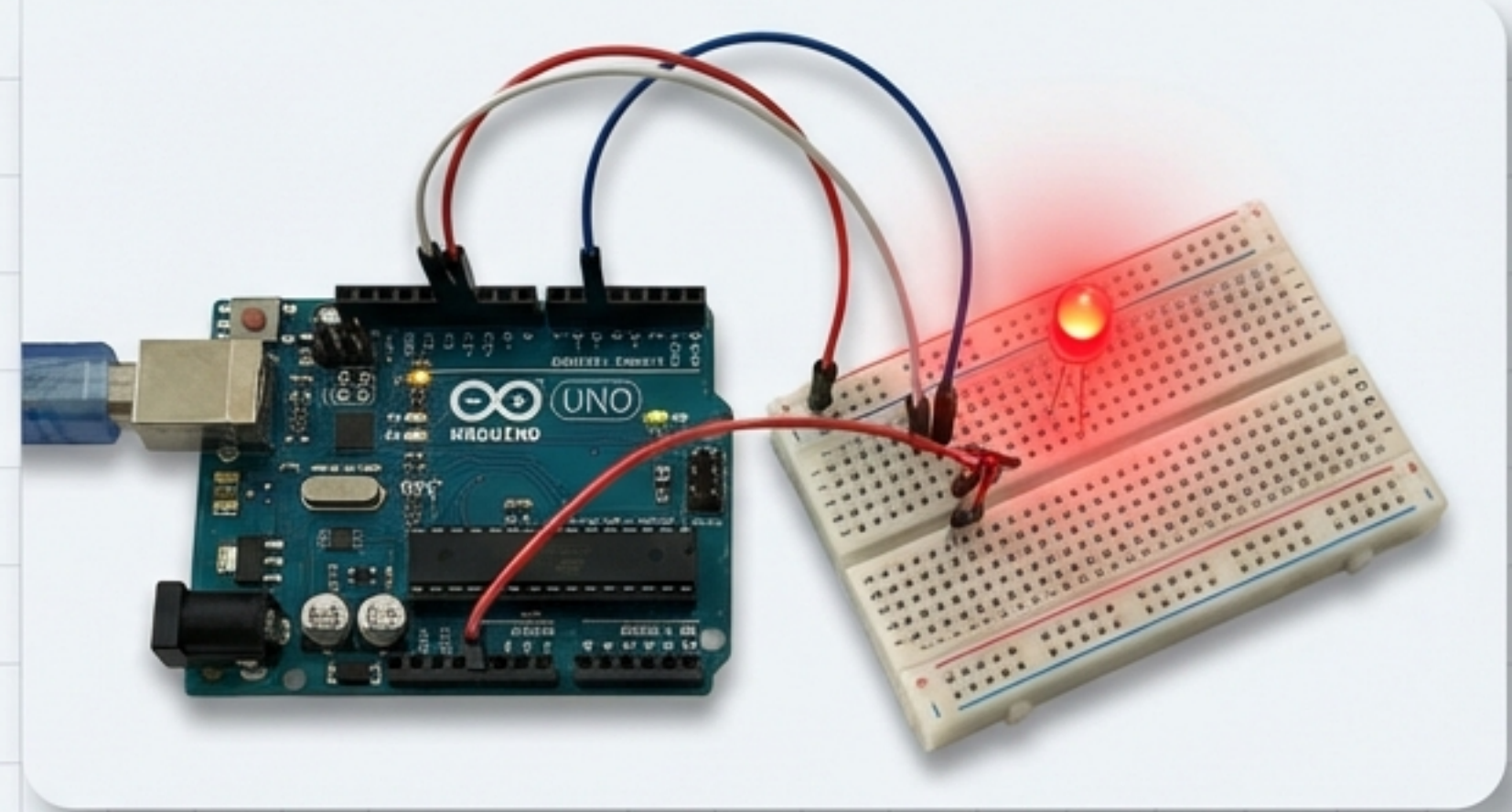
Action: LED ON

Action: LED OFF

Upload and Test



Click the **Upload** button.
Wait for the transfer to finish.



Result: The LED on your breadboard should be blinking!



Challenge: Change the "Wait" block to 0.5 seconds. What happens?

Real World Applications



Security Alarm:

Sensor detects motion -
-> Alarm sounds



Automatic Doors:

Sensor sees person ->
Motor opens door



Fire System:

Sensor detects smoke ->
Sprinkler activates

You are learning the same Input -> Process -> Output logic that runs the world!

Tech Flash: Key Terms



Arduino

The Brain. An open-source platform for building interactive devices.



Sensor

The Input. Converts physical changes into digital data.



mBlock

The Language. Block-based coding perfect for beginners.



USB Cable

The Lifeline. Provides power and data connection.